

Dye tracer visualization of flow patterns and pathways in glacial sandy till at a boreal forest hillslope



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ABSTRACT

Dyes are valuable tracers in visualizing flow patterns and pathways in soil. We applied the dye *Acid Blue 9* to unsaturated and saturated soil profiles at a boreal forest hillslope consisting of glacial sandy till, and determined the soil physical properties from soil samples. The objective was to characterize preferential flowpaths, investigate their porosity, extent and connectivity, and complement earlier findings on subsurface flow formation at the site. According to the results, preferential flowpaths were formed by roots, erosion related to soil water flow, freezing–thawing cycles, and soil fauna. The role of roots and stones in the formation of preferential flowpaths was emphasized. Porosity of preferential flowpaths was $5.1 \pm 1.8\%$, and they extended to a depth of about 55 cm from the soil surface; the deepest roots reached the same depth. When the soil saturated, individual preferential flowpaths self-organized into continuous and well-connected lateral flow systems along the slope. At the slope shoulder, preferential flow network covering the entire soil profile, as well as preferential flowpaths on the underlying bedrock surface, were considered crucial for runoff generation. In the midslope area, runoff generation was characterized by lateral preferential flow and the transmissivity feedback phenomenon. At the slope foot, preferential flowpaths in soil below the eluvial horizon were the major runoff contributors. A full saturation of the soil profile at the slope is unlikely under natural conditions. However, lateral flow was found to occur also in unsaturated soil.

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1. Introduction

Preferential flowpaths characterize soils at forested hillslopes (e.g., Uchida et al., 2005). They are networks of large pores (i.e. macropores), soil pipes and other void spaces in soil that are typically formed in forest soils by soil fauna, roots, erosion caused by water flow, and freezing–thawing phenomena (Aubertin, 1971; Beven and Germann, 1982, 2013; Koch et al., 2013). Preferential flow refers to all fast flow phenomena that are associated with a fraction of the total soil pore space, either accelerating or delaying the transport of dissolved matter, depending upon the location of the matter compared to the locations of the preferential flowpaths (Allaire et al., 2009). Preferential flow networks can remain stable for decades in forest soils (Hagedorn and Bundt, 2002), and both the vertical infiltration of water and solutes into soil, as well as their lateral movement along hillslopes towards streams and catchment outlets are affected by preferential flowpaths (e.g., Klaus et al., 2013). In addition to the abovementioned factors, swelling and shrinking processes can form preferential flowpaths in clay or organic soils; in agricultural fields, the efficiency of field drainage

relies on the existence of preferential flowpaths that can transport water and solutes rapidly from the soil surface to the subsurface drains (e.g., Alakukku et al., 2010).

Assessment of the type, extent and connectivity of preferential flowpaths, as well as the quantification of preferential discharge in forest hillslopes usually rely on tracer studies. While dye tracers have been used for visualizing the preferential infiltration into soil (e.g., Backnäs et al., 2012; Shougrakpam et al., 2010; Bogner et al., 2008) and the lateral preferential flow in hillslopes (e.g., Anderson et al., 2009; McGuire et al., 2007; Noguchi et al., 1999), ion and isotope tracers have been used for calculating the fraction of event vs. pre-event water, and the fraction of preferential vs. matrix flow in the total runoff from hillslopes (e.g., Laine-Kaulio et al., 2014a; McGuire et al., 2007; Lepistö et al., 1994). In agricultural sites, recent studies have included X-ray computed tomography to provide 3-D images of preferential flow networks in intact soil columns in laboratory conditions (e.g., Koestel and Larsbo, 2014; Luo et al., 2007; Mooney and Morris, 2004).

Studies on dye tracer visualization of preferential flow in forest soils have typically focused on fine-grained clay and loam soils (e.g., Wang and Zhang, 2011; Shougrakpam et al., 2010; Bogner et al., 2008), and characterization of preferential flowpaths in glacial tills with a coarser texture and high stone content has gained less attention. However, glacial tills are common in boreal forest environments in the Northern

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Hemisphere, and the main surface deposits in Fennoscandia (Beldring, 2002). Finnish soils, for instance, are dominated by Podzols with sandy till deposits (Yli-Halla and Mokma, 2002). 72% of forest land in southern Finland and 81% of forest land in northern Finland are on mineral soils of which 63% and 75%, respectively, are covered with medium to coarse textured glacial tills (Tomppo et al., 2011). Even though glacial till soils are generally considered to have a low hydraulic conductivity (e.g., Lind and Lundin, 1990), they may contain preferential flowpaths that significantly increase the subsurface water flow and solute transport velocities and amounts in forested hillslopes (e.g., Laine-Kaulio et al., 2014a).

According to Haldorsen and Krüger (1990), characteristics of tills that have the greatest influence on their hydrogeological properties are porosity, pore size distribution, macropore networks, heterogeneity and anisotropy; these properties are controlled by grain-size distribution, spatial distribution and orientation of soil particles, soil structural properties, and the degree of compaction of till. The soil structure of till has the strongest influence on the hydraulic conductivity, which decreases rapidly with depth, similarly to the volume of preferential flowpaths (e.g., Laine-Kaulio, 2011; Lind and Lundin, 1990). The higher volume of preferential flowpaths and the higher hydraulic conductivity of soil near the soil surface lead to a non-linear increase in lateral discharge when the soil saturates and water table rises to the highly conductive layers near the soil surface; this is called the transmissivity feedback phenomenon, and it characterizes runoff generation in many till hillslopes (e.g., Bishop, 1991). However, preferential flow in the soil material on the bedrock surface has been found the major source of runoff at hillslopes with very shallow (50–70 cm) till layers (e.g., Ilvesniemi et al., 2010).

Experimental evidence from field studies, especially in the context of lateral preferential flow, suggests that preferential flow processes vary considerably from site to site (Weiler and McDonnell, 2007). Characteristics of preferential flow networks, together with other soil properties, characteristics of the underlying bedrock, and climatic conditions, can be considered the key factors controlling the runoff generation at different sites but also at different locations of a same site. At the MaiMai experimental hillslope in New Zealand, preferential flowpaths have been identified to consist of soil pipes in the soil material on the bedrock surface at a depth of about 60 cm (e.g., McDonnell, 1990). At a forest site in South-East Germany, preferential flowpaths have been linked to rooting activity within a depth of about 1 m from the soil surface (e.g., Bogner

et al., 2010). A variable stone content characterizes our site, the forested hillslope in Kangaslampi, Finland, and is expected to affect the subsurface water flowpaths and patterns. The Kangaslampi slope has a glacial till soil cover above a low-permeable bedrock, and according to earlier studies including ion tracer experiments and solute transport modeling, runoff generation in the midslope area of the Kangaslampi site is characterized by preferential by-pass flow and the transmissivity feedback phenomenon (Laine-Kaulio et al., 2014a; Laine-Kaulio, 2011). The model results have clearly indicated that water flow and solute transport should be considered in two separate, but connected pore domains with strongly differing hydraulic properties at least in the uppermost 50 cm of the soil profile (Laine-Kaulio et al., 2014a).

In this study, we present a visualization of flow patterns and pathways in the midslope area of the Kangaslampi hillslope. We use the dye tracer *Acid Blue 9* to expose the preferential infiltration into unsaturated soil, as well as the lateral subsurface flow downslope near the water table under steady state conditions. In addition, we determine the porosity of preferential flowpaths at three elevation levels of the slope. The objective is to characterize preferential flowpaths in soil, investigate their porosity, extent and connectivity, and complement earlier findings on subsurface flow formation at the site. We utilize findings available from dye tracer studies at the slope shoulder and slope foot (Backnäs et al., 2015) and from ion tracer simulations in the midslope area (Laine-Kaulio et al., 2014a). The main hypothesis is that the dominant flow mechanisms behind the runoff generation in different parts of the hillslope are not the same because the soil profile thickness and the detailed characteristics of the preferential flow networks are different. Stones are expected to play a bigger role in the formation of preferential flowpaths than reported from other sites.

2. Material and methods

2.1. Study site

The Kangaslampi area in Eastern Finland (Fig. 1a) has been subject to long-term monitoring and a series of studies in forestry and hydrology (e.g., Finér et al., 1997; Laurén et al., 2005; Laine-Kaulio et al., 2014a, 2014b). The area belongs to the middle boreal forest zone. The long-term (1971–2000) mean annual air temperature is +1.9 °C, the mean annual maximum of soil frost depth is 22 cm, and the mean annual

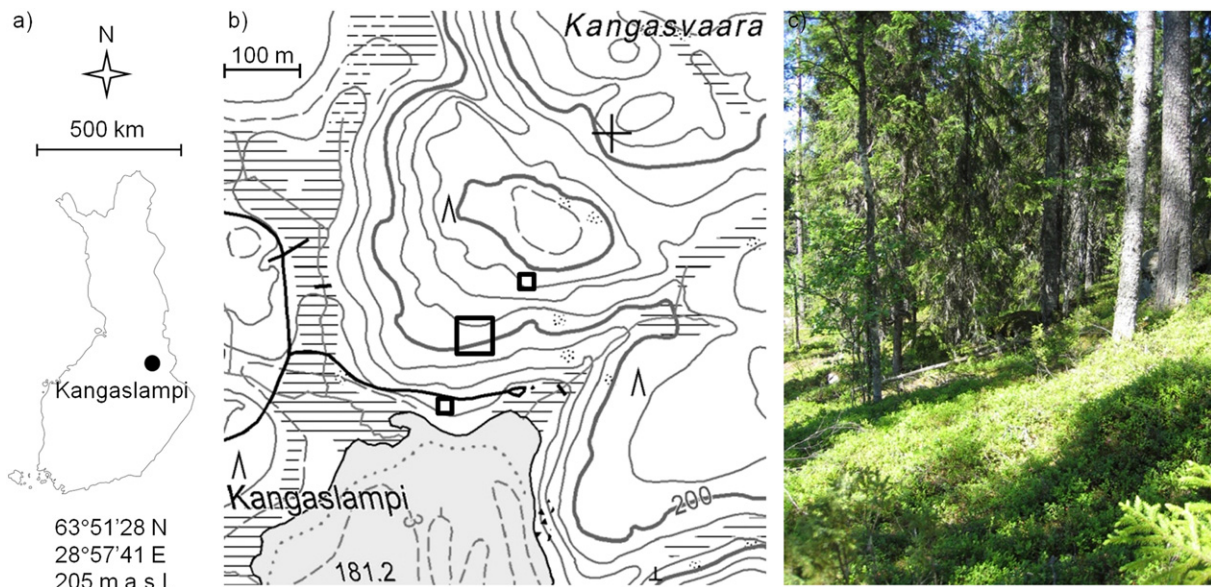


Fig. 1. Location of the Kangaslampi study area in Finland (a), locations of the slope foot, midslope and slope shoulder study plots in the experimental hillslope (for the slope portion terms see, e.g., Miyazaki, 2006) (b), and view on the midslope area (c).

precipitation is 564 mm with approximately 200 mm in the form of snow (Piirainen et al., 2007; SYKE, 2003). About half of the annual precipitation generates runoff, and the main yearly runoff event is induced by spring snowmelt. Topography is characterized by gently sloping hills that are surrounded by riparian areas, ponds and small lakes.

The present study concentrated on the midslope area of a 300 m long hillslope on the north side of the Kangaslampi pond (Fig. 1b). Soil sampling for analyzing the soil physical properties was performed in September 2005, and dye tracer experiments in June 2006. In addition, soil sampling for determining the soil physical properties at the slope shoulder and slope foot study plots (Fig. 1b) was carried out in June 2007.

The field layer vegetation of the site consisted of dwarf shrubs (*Vaccinium vitis-idaea* L., *V. myrtillus* L., and *Empetrum nigrum*), and feather mosses (*Pleurozium schreberi*) dominated the bottom layer (Fig. 1c). The mixed, coniferous forest was approximately 70 years old and composed of Norway spruce (*Picea abies* Karsten, 55%), Scots pine (*Pinus sylvestris* L., 28%) and white birch (*Betula pubescens* Ehrh., 17%) (Fig. 1c). Based on a relascope sample plot (Bitterlich, 1984), the mean tree height was 20 m and the mean volume 273 m³ ha⁻¹. Prior to the experiments, the mineral soil column had not been treated or disturbed by forestry practices; the tree stand had regularly been thinned but had no detectable impact on the underlying mineral soil column.

The dominant soil type was a Haplic Podzol with sandy till as the parent material (FAO, 1988); the mineral soil profile consisted of eluvial (E), illuvial (B), transitional (BC), and subsoil (C) horizons (Table 1). The soil was rather stony, with the highest mean stone content of 34% near the soil surface in the midslope area (Table 1). The underlying bedrock was gneiss granite and granodiorite (Korsman et al., 1997), and the latest stage of bedrock erosion and deposition of glacial till had occurred during the Late Weichselian glaciations about 11500 years ago (Lunkka et al., 2004).

In the midslope area, the mineral soil profile was 69–116 cm thick, and it was overlain by an organic litter and mor humus layer (O) that had a mean thickness of 10 cm. The B horizon was partially cemented due to the formation of Ortstein by illuviated sesquioxides and organic matter. The mean slope angles of the soil and bedrock surfaces were 15% and 18%, respectively. At the slope shoulder and slope foot, thicknesses of the mineral soil profile were 40 cm and 50 cm, respectively, and the mean thickness of the O horizon was 15 cm. The mean slope angles of the soil and bedrock surfaces were 19% and 20%, respectively, at the slope shoulder study plot, and 18% and 7% at the slope foot study plot.

2.2. Determination of the porosity of preferential flowpaths

Porosity estimates of preferential flowpaths were determined from undisturbed soil core samples (number of samples 21, height 1.5–

7.0 cm, diameter 3.8–5.4 cm) that were taken from sampling pits at three elevation levels at the slope (Fig. 1b). The samples were saturated in a laboratory from below by placing them in boiled and cooled off water. The water level was at the level of the soil surface; as the soil core cylinders were higher, no water entered the samples from above. The volume of water that drained from the saturated samples by the force of gravity for 24 h was measured. This volume is called the *air capacity*, and it provides an estimate of the macropore volume of soil (Burger, 1922 after Germann and Beven, 1981). The samples were then saturated again for water retention measurements with a pressure plate apparatus; the difference between saturated water volume and water volume corresponding to different pressure heads provided further estimates of the porosity of preferential flowpaths. We used the following equation:

$$h = \frac{30}{d} \quad (1)$$

where h is the hydraulic pressure head [m] and d is the threshold diameter for a macropore [μm] (Vakkilainen, 1986). We chose to use a threshold diameter of 300 μm , relying on the finding that pores larger than 300 μm in equivalent cylindrical diameter allow rapid non-equilibrium flow in soil (Jarvis, 2007). We thereby estimated the porosity related to preferential flowpaths as the difference between the water volume fraction in soil cores at saturation and at 10 cm suction. In order to estimate the porosity of preferential flowpaths at a scale larger than the small soil core samples, the porosity results were corrected by the stone content data (Table 1).

2.3. Dye tracer experiments

Acid Blue 9, also known as Brilliant Blue FCF ($\text{C}_{37}\text{H}_{34}\text{N}_2\text{Na}_2\text{O}_9\text{S}_3$), has good visibility and low retardation in soil at the same time, and it is not toxic to the environment (Flury and Flühler, 1994; Weiler, 2001; Weiler and Flühler, 2004). It was therefore chosen for the dye tracer experiments. In the first two experiments, dye solution was poured on the surface of unsaturated soil profiles in the midslope area, staining the infiltration into soil. In the third experiment, dye solution was poured into saturated soil, staining the lateral flow downslope near the water table in soil in the midslope area. The total dye load in each experiment was chosen so that the remaining dye concentration in soil did not exceed the limit recommended by Flury and Flühler (1994), i.e., 1 mg l⁻¹.

2.3.1. Experiments in unsaturated soil

The first two experiments were performed in the same way: 10 l of 0.5 g l⁻¹ dye solution was poured slowly on an area of 50 × 50 cm² with a sprinkle (Figs. 2a and 3a). After this, 3 l of tracer-free water was

Table 1
Soil physical properties at the Kangaslampi hillslope.

Soil horizon	Depth [cm]	Parent material	Stone content [m ³ m ⁻³]	Porosity [m ³ m ⁻³]	Porosity excl. stones [m ³ m ⁻³]	OM content [m ³ m ⁻³]	Dry unit weight [kN m ⁻³]	Initial degree of saturation [–]
<i>Slope shoulder</i>								
E	15–20	Sand	0.05	0.48	0.46	0.009	13.0	0.39
B	20–37	Sandy till	0.12	0.61	0.54	0.057	10.5	0.39
BC	37–55	Coarse sandy till	0.03	0.48	0.47	0.021	15.0	0.54
<i>Midslope</i>								
E	10–19	Sandy till	0.34	0.49	0.32	0.014	11.8	0.29
B	19–33	Sandy till	0.30	0.47	0.33	0.067	15.5	0.42
BC	33–50	Sandy till	0.17	0.40	0.33	0.013	14.4	0.30
C	50–102	Sandy till	0.09	0.34	0.31	0.005	15.3	0.36
<i>Slope foot</i>								
E	15–27	Sandy till	0.30	0.42	0.29	0.010	12.4	0.75
B	27–45	Sandy till	0.18	0.57	0.47	0.090	9.6	0.89
BC	45–52	Sandy till	0.10	0.53	0.48	0.022	10.5	0.86
C	52–65	Sandy till	0.05	0.50	0.48	0.011	14.3	0.46

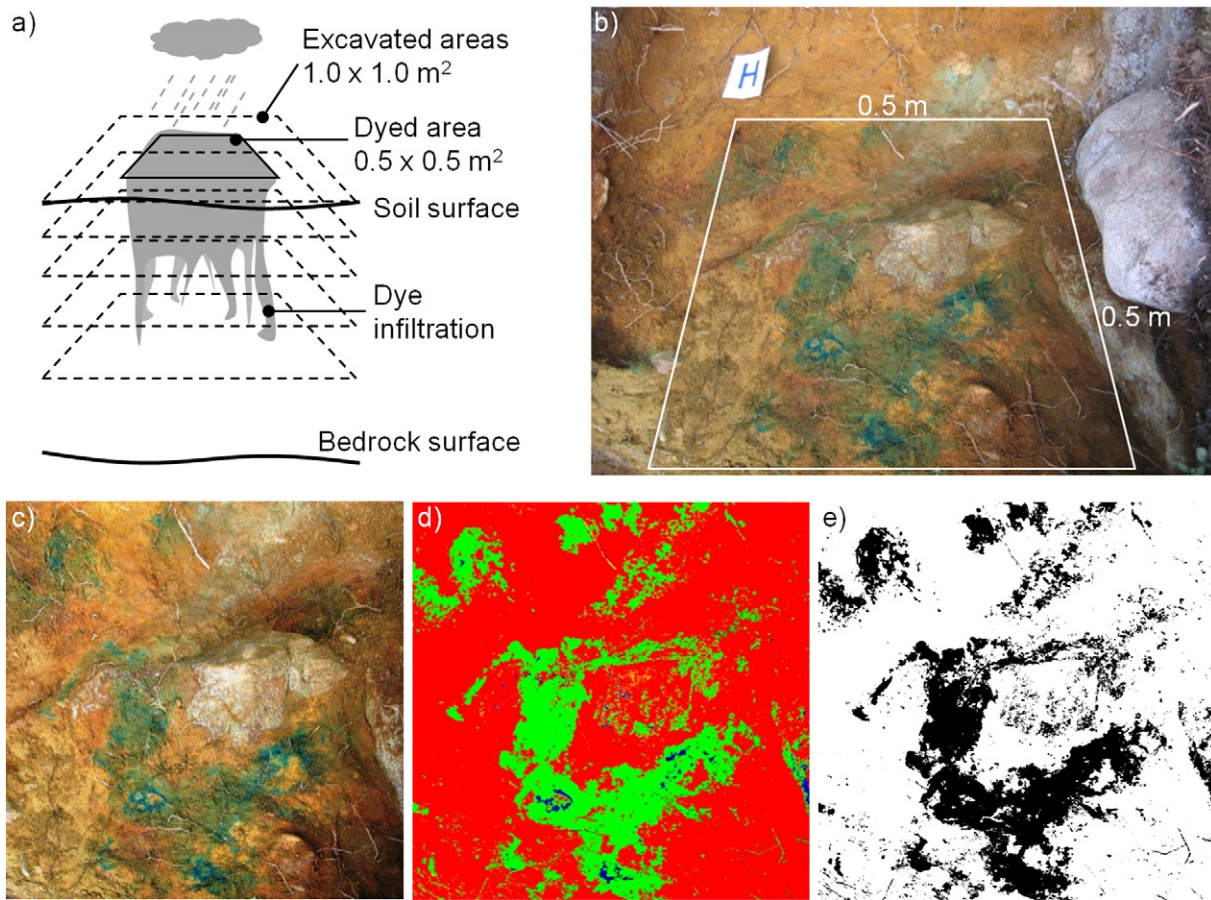


Fig. 2. Dye infiltration experiment when excavating the soil profile in horizontal cross-sections (a), photograph of a cross-section at the depth of 25 cm from the top of the mineral soil profile (b), and its delimitation and perspective correction (c), red, green and blue conversion (d), as well as black and white conversion (e).

poured onto the plot to wash down the dye from the ground vegetation. Flury and Flühler (1994) recommended higher dye concentrations, i.e. 3–5 g l⁻¹, for similar experiments to ensure good visibility of the dye in soil. The smaller concentration used in our study was enough to produce a distinctly visible dye pattern in soil with the total dye solution load of 10 l. The dye application resulted in blue to green-colored patches in the gray to yellow and red-toned Podzol profile (e.g., Figs. 2b, 3b and 4b).

The soil was dry before the experiments (Table 1). The 13 l of irrigation corresponded to 28% of the empty pore volume of soil, and about 48% of the empty pore volume of the O, E, B, and BC horizons where dye was found (cf. Section 3.2). Before starting the excavations, the O horizon was carefully removed. The excavated areas were 1.0 × 1.0 m², i.e., 25 cm wider in each direction than the area where the dye was applied to (Figs. 2a and 3a). Dye infiltration into the first plot was explored by excavating horizontal cross-sections one at a time from topsoil down towards the bedrock (Fig. 2a). Spades, hoes and brushes of different sizes were used in the excavations. The second plot was excavated in vertical cross-sections (Fig. 3a). The horizontal cross-sections were used for determining the dye coverage on areas normal to the vertical infiltration direction at 5 cm depth intervals, and the vertical cross-sections were used for measuring the infiltration depths of the dye from the top of the mineral soil profile, as well as the dye coverage of each soil horizon; types of preferential flowpaths were investigated from both the horizontal and the vertical cross-sections.

To determine the dye coverage from a cross-section photograph, a perspective correction was performed if the photo was not taken perpendicularly to the cross-section area, and the photo was cropped to the area where the dye was applied to (Figs. 2b–c and 3b–c). The blue

and green dye patches were distinguished from undyed soil material by converting the colors of the photos to red, green and blue (Figs. 2d and 3d). In addition, black and white images were created so that red (undyed) areas were presented in white, and green and blue (dyed) areas were presented in black (Figs. 2e and 3e). The free GNU image manipulation program, Gimp 2.8, was used for the photo processing and determinations of the dye coverage at different depths and in different soil horizons.

2.3.2. Experiments in saturated soil

The third experiment was conducted within a field of observation wells that was used a year earlier for hillslope-section-scale measurements of lateral chloride transport within the midslope area (Laine-Kaulio et al., 2014a; Laine-Kaulio, 2011). First, a 3.6 m long perforated irrigation tube was placed upslope from the observation well field, and 1 m³ of tracer-free water was irrigated to the soil a day before the dye tracer experiment through the line-type irrigation source to increase the initial soil moisture content. During the initial irrigation, water table rose up to the E horizon within the distance of about 2.0 m from the irrigation source. When the initial irrigation was stopped, the water table fell. No water table was observed in the observation wells above the bedrock when the dye tracer experiment was started the next day.

The actual dye tracer experiment was started by irrigating the soil with tracer-free water until the soil had saturated up to the O horizon at a width of at least 2.0 m and a length of at least 3.0 m downslope from the irrigation source. Then, the irrigation was stopped. While the water table withdrew down, a 1 m wide and 10–15 cm deep trench was dug at a distance of 60 cm from the irrigation tube. The bottom of

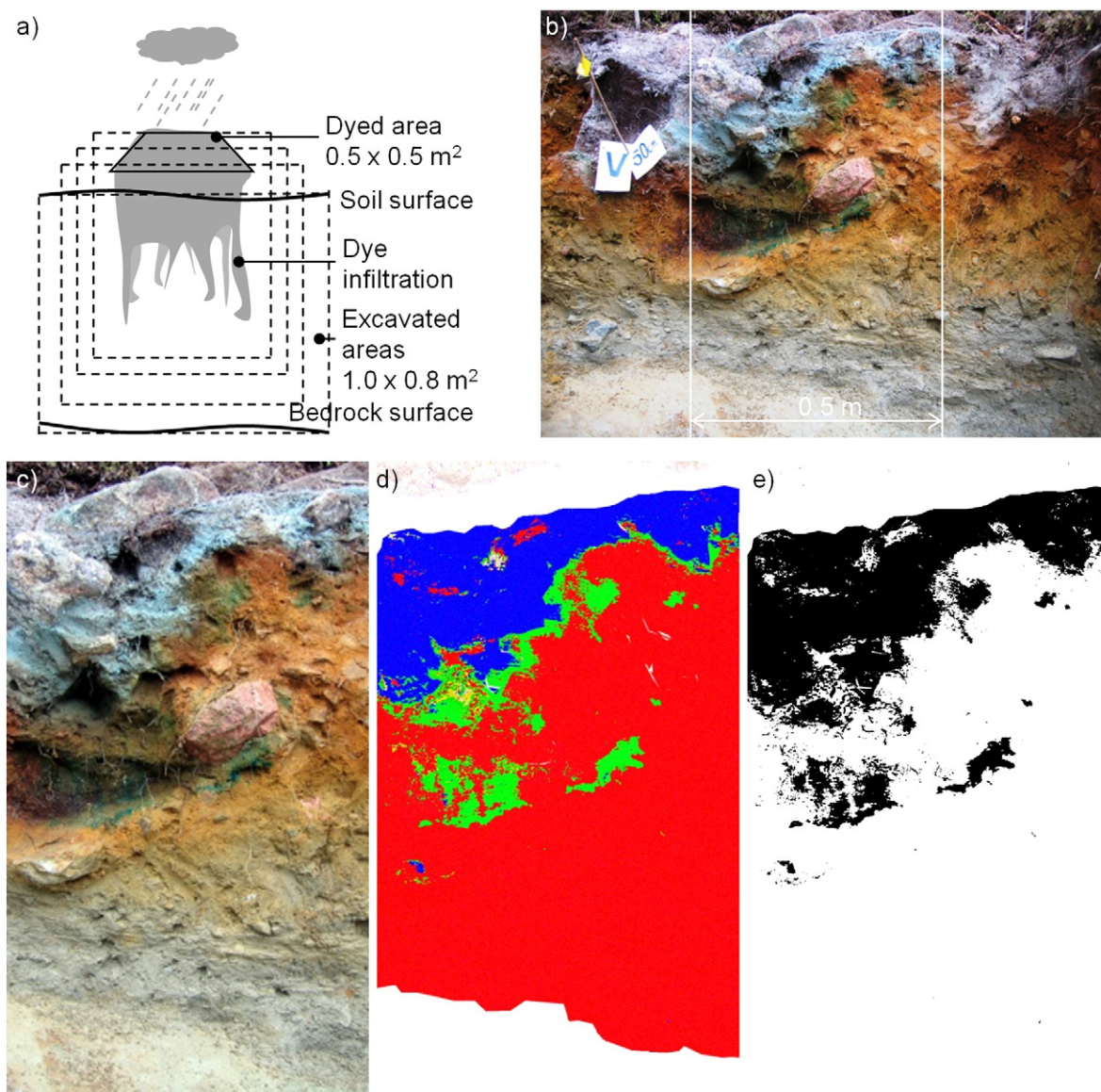


Fig. 3. Dye infiltration experiment when excavating the soil profile in vertical cross-sections (a), photograph of a cross-section in the middle of the plot, at the distance of 50 cm from the front wall of the soil profile (b), and its delimitation (c), red, green and blue conversion (d), as well as black and white conversion (e).

the trench was in the E horizon, a few centimeters down from the interface to the O horizon. When the water table had fallen down to the B horizon in the observation wells, 10 l of 1.0 g l^{-1} dye solution were poured slowly into the trench (Fig. 4a). Thus, a 1 m wide dye pulse was sent downslope above the water table that was located in the B horizon.

The water level was chosen to be tuned to the B horizon in this experiment because the B horizon had been found to be the lowest soil horizon where preferential flowpaths were clearly present (cf. Section 3.2; Laine-Kaulio et al., 2014a; Laine-Kaulio, 2011). Immediately after pouring the dye solution to the trench, additional 10 l of tracer-free water was poured into the trench to wash away the dye that remained in the trench surfaces. In addition, irrigation was started again after sending the pulse downslope to feed the fast lateral flow in the hillslope. The irrigation was ended after 5 min, and the slope was left to drain overnight. Water table in the observation wells did not rise during the 5 min irrigation, and the observation wells were empty when the study area was excavated into vertical cross-sections the next day. Photographs of the cross-sections were processed in the same way as the photos from the experiments in unsaturated soil (Fig. 4b–d), and the

dye coverage at different depths, as well as the horizontal width of the dye accumulation near the water table in the B horizon were determined.

3. Results

3.1. Porosity of preferential flowpaths

Measured values of the porosities of preferential flowpaths are presented in Table 2. In addition to the measured values, Table 2 shows the calibrated porosities of preferential flowpaths that originate from the modeling study of Laine-Kaulio et al. (2014a) in the midslope area. The porosities related to pores larger than $300 \mu\text{m}$ in diameter were systematically larger than the air capacity values; the only exception was the sample taken from the C horizon. This sample was the only one representing the subsoil clearly below the depth that the preferential flowpaths reached (cf. Section 3.2). Based on all measurements, the mean air capacity in the E, B and BC horizons was 3.6%, and the mean porosity related to pores larger than $300 \mu\text{m}$ in the E, B and BC horizons was 5.1%. In the midslope area, the mean porosity of preferential

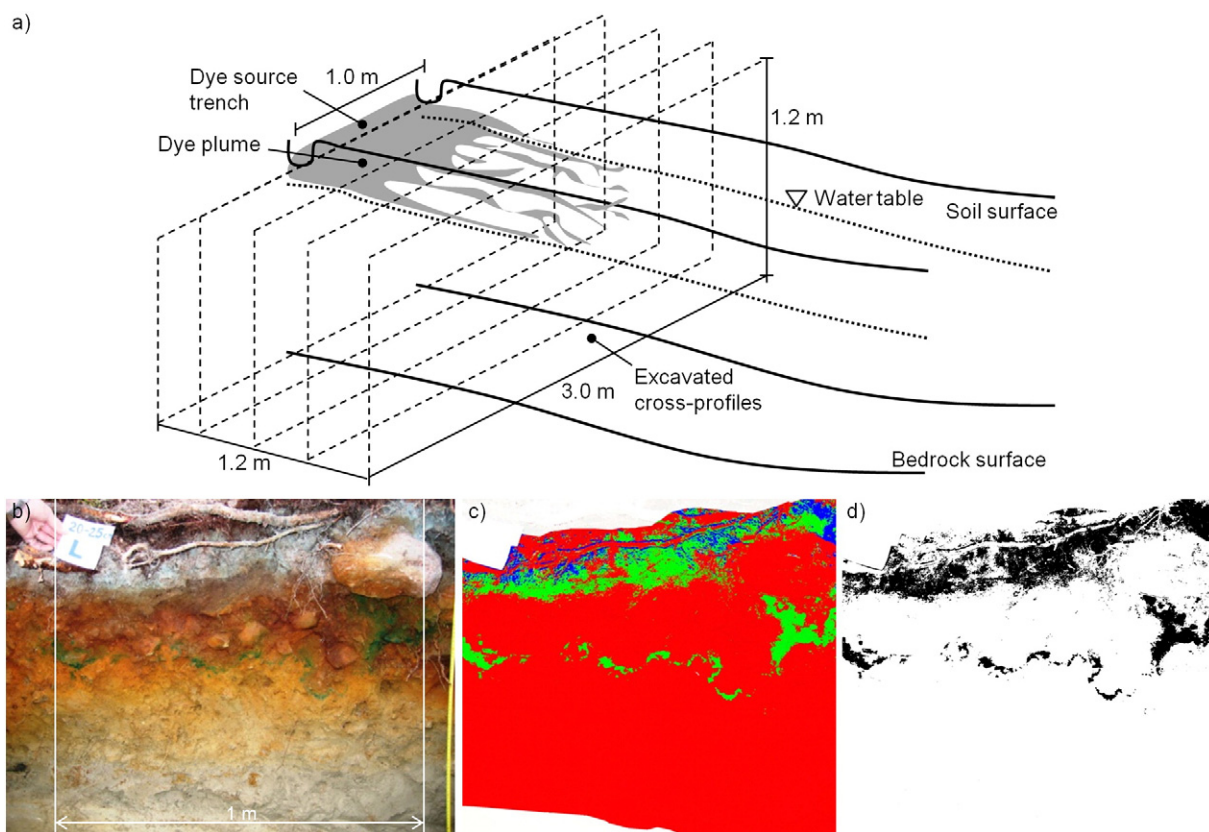


Fig. 4. Lateral dye transport experiment when sending a 1.0 m wide dye pulse downslope above the water table that was located in the B horizon (a), photograph of a vertical cross-section at the distance of about 20 cm from the dye source trench, and its delimitation limits (b), its red, green and blue conversion (c), and black and white conversion (d).

flowpaths reduced with depth, being highest near the soil surface, while at the slope foot study plot, the highest value originated from the B horizon, and at the slope foot study plot from the B horizon or the BC horizon (Table 2).

The representativeness of the measured values was weakened by the small soil core sizes and the rather small number (≤ 3) of cores available for each soil horizon at each study plot, combined with the observed high variation in the results (Table 2). Compared with the calibrated values of Laine-Kaulio et al. (2014a) for the midslope area, the mean porosities related to pores larger than 300 μm were in the same

magnitude and the mean air capacity values were lower. Because the calibrated estimates of the porosities of preferential flowpaths represent the scale of a hillslope section and originate from a successful simulation of chloride transport at different depths in soil according to a detailed 3-D dual-permeability model, the calibrated values were considered most reliable. However, the calibrated values only applied to the midslope area.

3.2. Dye patterns in unsaturated soil

Dye coverage in relation to soil depth, as determined from the top of the mineral soil profile in both the horizontally and the vertically excavated plots, is presented in Fig. 5. The vertical distribution of dye decreased strongly with depth and reflected the pattern of the midslope porosity variables in Table 2. Within the depth of 0–10 cm and 20–30 cm, the dye coverage was higher in the horizontal cross-section than on the vertical cross-sections on average, while within the depth of 10–20 cm and 30–50 cm, the dye coverage was lower in the horizontal cross-section than on the vertical cross-sections on average. Differences in the amount of dye between different cross-sections are due to the heterogeneity of the soil.

The depth distribution of the dye followed the soil horizon thicknesses so that where the E and B horizons were thicker, the dye was found deeper (Figs. 3c and 5). As for the thickness of the E horizon and B horizon, stones increased the depth that these horizons reached (Figs. 3c and 6a). The maximum depth of the E horizon was 24 cm and the maximum depth of the B horizon was 59 cm from the top of the mineral soil profile. The dye patches reached down to the BC horizon and no dye was detected in the C horizon (e.g., Fig. 3c). It is important to notice that the interfaces between soil horizons are presented as mean values in Fig. 5. Thus, even though the dye coverage exceeds 0% in the

Table 2

Average measured air capacity, θ_{ac} , porosity related to soil pores with a diameter larger than 300 μm , $\theta_{300 \mu\text{m}}$, and porosity of the preferential flow domain of soil as simulated with a dual-permeability model (Laine-Kaulio et al., 2014a), θ_{psim} (measured variation, sample size).^a

Soil horizon	θ_{ac} [$\text{m}^3 \text{m}^{-3}$]	$\theta_{300 \mu\text{m}}$ [$\text{m}^3 \text{m}^{-3}$]	θ_{psim} [$\text{m}^3 \text{m}^{-3}$]
<i>Slope shoulder</i>			
E	0.041 (0.026–0.067, 3)	0.052 (0.048–0.061, 3)	
B	0.050 (0.034–0.067, 2)	0.075 (0.021–0.128, 2)	
BC	0.041 (0.037–0.046, 2)	0.064 (0.054–0.073, 2)	
<i>Midslope</i>			
E	0.042 (0.019–0.058, 3)	0.056 (0.034–0.079, 2)	0.070
B	0.025 (0.020–0.031, 2)	0.050 (0.036–0.064, 2)	0.035
BC	0.020 (–, 1)	0.044 (–, 1)	0.020
C	0.013 (–, 1)	0.012 (–, 1)	0.015
<i>Slope foot</i>			
E	0.022 (0.014–0.030, 2)	0.033 (0.017–0.049, 2)	
B	0.055 (0.037–0.073, 2)	0.061 (–, 1)	
BC	0.044 (0.024–0.065, 2)	0.074 (0.037–0.111, 2)	
C	0.041 (–, 1)	0.044 (–, 1)	

^a All values have been corrected by the stone content data (cf. Table 1).

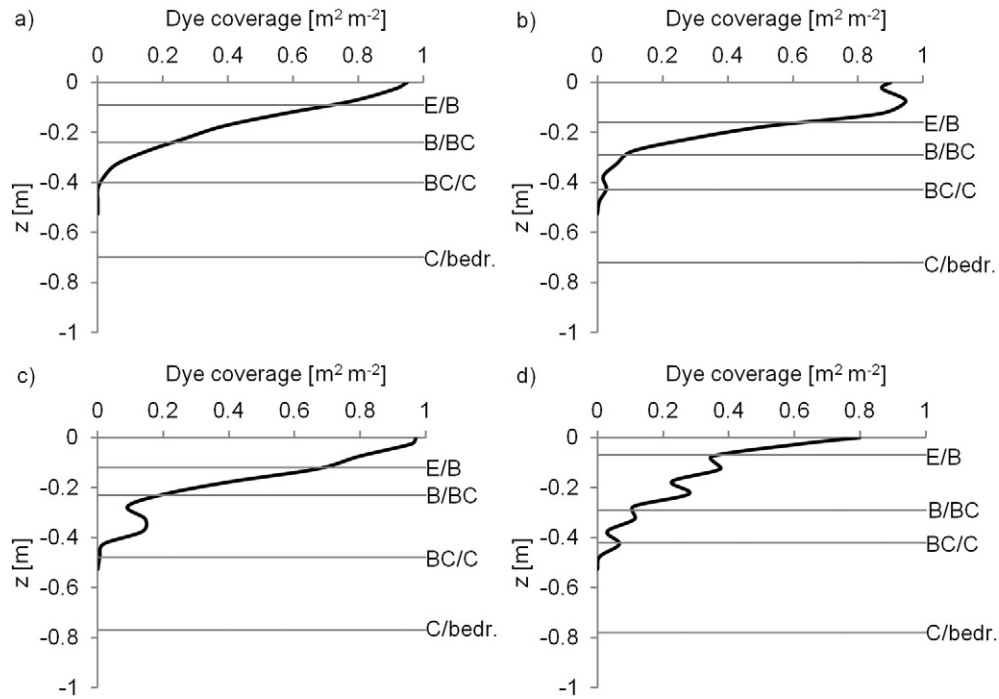


Fig. 5. Dye coverage in relation to soil depth from the top of the mineral soil profile and average depths of the boundaries between soil horizons in the horizontally excavated plot (a) (cf. Fig. 2a) and in the vertically excavated plot (cf. Fig. 3a) at the distance of 35 cm (b), 50 cm (c) and 65 cm (d) from the front wall of the soil profile.

C horizon in Fig. 5b and d, there was no dye in the C horizon on these cross-sections, but the thickness of the upper soil horizons exceeded the presented mean values at the locations of the deepest dye infiltration.

The maximum depth that the dye reached was 45 cm from the top of the E horizon and 55 cm from the soil surface including the O horizon. Roots were found to reach the same depth as the dye.

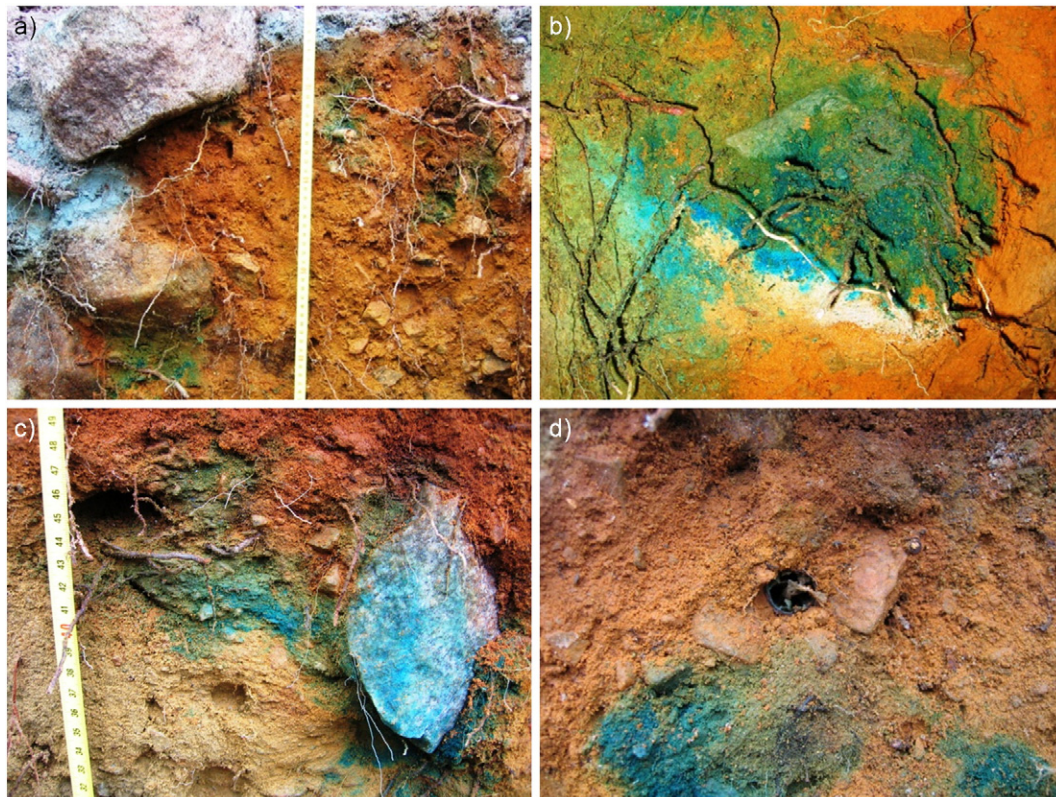


Fig. 6. Difference between the dye coverage in the E horizon and B horizon (a), dye accumulation on and around live roots in the B horizon (b), dye accumulation on and around a stone in the B-BC horizon (c), and a 1 cm wide decayed root hole in the B horizon with blue dye inside and below (d).

The dye spread most uniformly in the loose E horizon (Figs. 3c and 6b). For instance on the vertical cross-section at the 50 cm distance (i.e., on the middlemost cross-section), 92% of the E horizon was dyed (Fig. 3c). The dye coverage reduced rapidly in the B horizon, and the dye covered about half of this soil horizon on average. On the vertical cross-section at the 50 cm distance, 53% of the B horizon was colored (Fig. 3c). The boundaries of the dye patches were clear (e.g., Fig. 2c). In the BC horizon, the darkness of the dye started to fade, and the dye covered 13% on the vertical cross-section at the 50 cm distance (Fig. 3c).

The dye was found to accumulate around and, in particular, underneath stones (Figs. 2c, 6a and c). All stones that were found in the E horizon in the horizontal excavations had a dark blue hue underneath. Also roots were found throughout the E horizon. In the B horizon, half of the stones that were removed in the horizontal excavations, had dye accumulated below. Some of these stones even had distinct flow stripes on their surfaces or the surfaces were completely covered with dye (Fig. 6c). Even though roots were found to reach the same depth in soil as the dye, and the dye and roots both accumulated near stones, surfaces of living roots were not systematically colored; dye was found on and around some roots (Fig. 6a–c). One large decayed root hole, distinguishable with a bare eye, and with traces of dye inside, was detected in the vertical excavations at the boundary between the B horizon and BC horizon (Fig. 6d).

3.3. Dye patterns in saturated soil

The excavations were started at a distance of 2.5 m downslope from the dye source trench (cf. Fig. 4a). Because no dye was detected at this distance, new profiles were exposed in intervals of 20–30 cm by proceeding gradually upslope towards the trench. Small, faint, barely visible traces of the dye were found at a distance of 1.5 m. Fig. 7 shows the dye coverage in relation to depth on vertical cross-sections (normal to the lateral flow direction) at the distances of 0, 0.25, 0.40, 0.60, 0.90 and 1.20 m downslope from the 1 m wide dye source trench (cf. Fig. 4a). The dye patterns in highly saturated soil (Fig. 4b) are similar to those in unsaturated soil (Fig. 3b); the main difference is that in wet soil the spreading and continuity of the dye pattern is higher, hiding many of the individual flowpaths on roots or stones that could be detected in unsaturated soil.

Dye detected in the E horizon at all distances was a sign of lateral flow above the water table in highly but not fully saturated soil. Based on the soil moisture data (Laine-Kaulio, 2011), the average degree of saturation in the E horizon was 92% when the water table had fallen to the B horizon and when the dye pulse was sent downslope. The highest dye coverage was detected in the E horizon at the distances of 0.2, 0.4 and 0.6 m (Fig. 7b–d), and in the B horizon at the distances of 0, 0.9 and 1.2 m (Fig. 7a, e–f). The mean thickness of the E horizon had a clear impact on the dye patterns: at cross-sections with a high mean thickness of the E horizon, the maximum dye coverage was high compared to the cross-sections with a low mean thickness of the E horizon (cf. Fig. 7b, d and f with Fig. 7a, c and e). Even though the dye coverage was higher within the E horizon than within the B horizon at three cross-sections, dye was found to accumulate as a clear, thin, tortuous layer near the water table in the B horizon at all distances; an example of this is shown in Fig. 4b–d. Fig. 8 shows the horizontal continuity of the dye accumulation (cf. Fig. 4b–d) near the water table in the B horizon. The horizontal spread of the dye, normal to the flow direction, increased until the distance of about 0.9 m from the dye source trench, and then sharply decreased.

3.4. Comparison to lateral chloride transport

To provide a direct comparison to dual-permeability simulations of lateral chloride transport in the midslope area (Laine-Kaulio et al., 2014a), Fig. 9 shows as an example the relation between the dye coverage (as determined from the middlemost cross-section of the vertically

excavated plot) and three different characteristics of lateral preferential flow. While the measured air capacities and the porosities related to pores larger than 300 μm in diameter do not correlate with the dye coverage, the calibrated porosities of the preferential flow domain show an exponential increase in relation to the dye coverage (coefficient of determination, $R^2 = 1.00$) (Fig. 9a). The saturated hydraulic conductivity of the preferential flow domain, as calibrated with the dual-permeability model, and the dye coverage follow a second order polynomial relation ($R^2 = 1.00$) (Fig. 9b). The simulated lateral chloride transport in saturated soil and the dye coverage are linearly correlated ($R^2 = 0.99$) (Fig. 9c).

4. Discussion

4.1. Characterization of preferential flowpaths

Beven and Germann (1982) grouped macropores into pores formed by soil fauna, pores formed by plant roots, cracks and fissures, and natural soil pipes. Reflected against this grouping, our dye tracer experiments revealed natural soil pipes and pores formed by plant roots. Natural soil pipes have been found to form because of erosive actions of subsurface flows (Zaslavsky and Kasiff, 1965 after Beven and Germann, 1982). We discovered dyed stone surfaces in all soil horizons where dye was found, and a close look at the stone surfaces inside the soil revealed clearly visible voids around the stones. The voids around, and specifically below stones were considered to result from subsurface erosion, combined with the effect of soil frost on soil around stones. The maximum depth of soil frost in forested till recorded in the area 1971–2000 is 84 cm; the mean annual maximum is only 22 cm because an insulating layer of snow usually covers the soil early in the winter in Eastern Finland (SYKE, 2003; Soveri and Varjo, 1977).

The decayed root hole with blue dye inside (Fig. 6d) was a clear indication of a pore formed by plant roots. There was a visible bark around the root hole, showing that barks of tree roots may resist decay longer than the xylem, and a hose type of a macropore can be formed (Gaiser, 1952 and Aubertin, 1971 after Beven and Germann, 1982). The role of living roots in the formation of preferential flowpaths was not fully unambiguous in our experiments because no systematic dye accumulation was found in the vicinity of living roots. Dye was found on and around some roots, the amount of roots as well as the dye coverage reduced with depth, and both the roots and the dye patterns extended down to the same depth (i.e., to 45 cm from the top of the mineral soil profile). In addition, roots were found especially below stones, where the dye, as well as water and nutrients accumulate (cf. Backnäs et al., 2012; Bundt et al., 2001).

At the slope shoulder and slope foot study plots, soil was less dense and less stony than in the midslope area (Tables 1 and 2), and the role of root channels as preferential flowpaths has been found more evident. Backnäs et al. (2015), for instance, made dye tracer experiments similar to the present study to distinguish preferential flowpaths from the soil matrix and to investigate the sorption of phosphorous in these two pore domains. They found that tree roots were systematically colored in the slope shoulder and slope foot study plots, and at the slope shoulder where the soil profile was most shallow, the roots and the dye reached the bedrock surface. In a Norway Spruce forest site in South-East Germany, Bogner et al. (2010) noted that roots constituted the main preferential flowpaths and induced macropore flow in the soils characterized as Haplic Podzols with a sandy to loamy texture and a low stone content of less than 5%. The root density was also found to be clearly higher in the preferential flowpaths than in the soil matrix at this German site. However, one of the three plots investigated by Bogner et al. (2010) had a high stone content of 10–50% in the Bw horizon at a depth of 33–55 cm, and at this particular plot the dye coverage was highest in the stony soil layer.

In general, decayed and living roots have been found to form networks of relatively large, continuous, interconnected, and open or

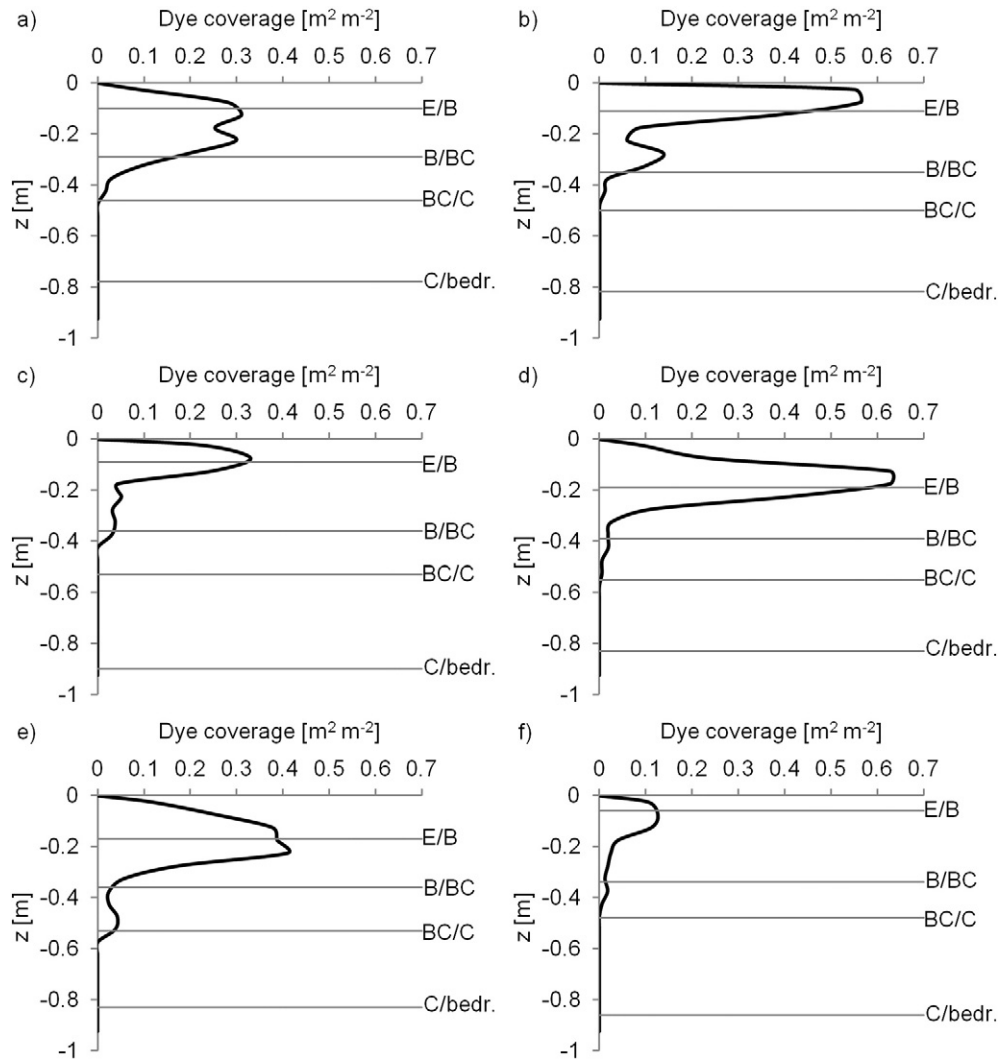


Fig. 7. Dye coverage in relation to soil depth from the top of the mineral soil profile at the distance of 0 m (a), 0.2 m (b), 0.4 m (c), 0.6 m (d), 0.9 m (e) and 1.2 m (f) from the dye source trench in the lateral dye transport experiment.

partially filled channels in forest soils, serving as preferential pathways (Aubertin, 1971). Roots affect the soil structure because the entrance and expansion of a root in soil compresses the soil adjacent to it and

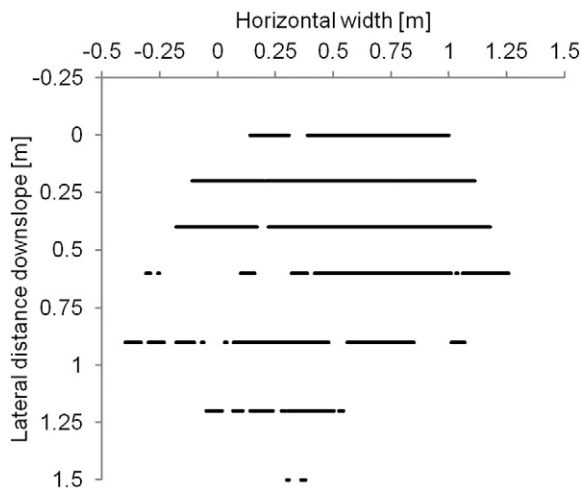


Fig. 8. Dye accumulation near the water table in the B horizon at different distances downslope from the dye source trench.

locally changes the porosity and bulk density (Aubertin, 1971). When roots decay, spots of lower density and higher porosity, or even a clear soil pipe remains in soil. According to Aubertin (1971), channels related to decayed root holes may comprise up to at least 35% of the near surface soil volume, with a rapid decrease with depth. Beven and Germann (1982) have noted that the distinction between pores formed by live and decayed root holes may be hard to make, as there is a tendency for new roots to follow the channels of previous roots. Backnäs et al. (2012) suggested that in forested heterogeneous glacial till soils, living roots act as contributors to preferential flow by providing connections from stone to stone and from stone to coarse grained spots in soil, enabling the formation of continuous preferential flow networks.

Combining all available observations from the Kangaslampi hillslope to the above presented, earlier findings made by others, we inferred that roots significantly affected the formation of preferential flowpaths at our forest site, together with soil formation under erosion caused by soil water flow and freezing–thawing phenomena. Roots, subsurface water flow and soil frost all alter the density and pore-size distribution of soil. As roots accumulate in the vicinity of stones, where also water and solutes accumulate, preferential flowpaths related to roots are connected to the voids around stones. In addition, soil fauna may have had an effect on the flowpath formation at our site. Earthworms are the most conspicuous group of animals in most forested soils (Aubertin,

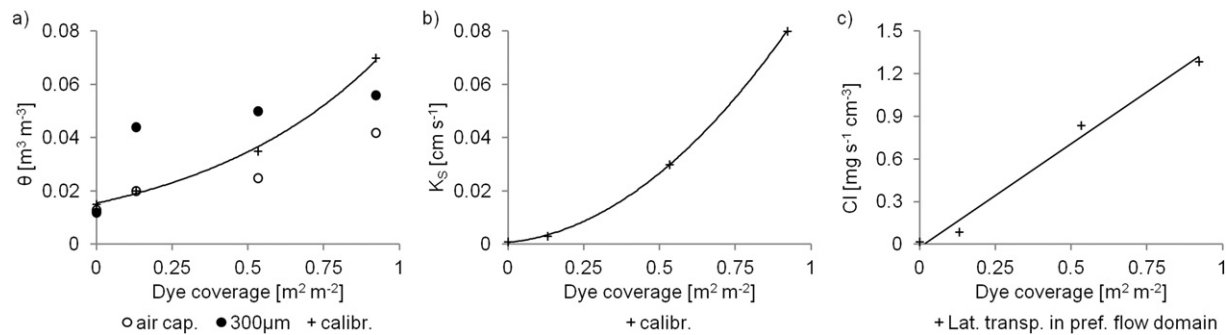


Fig. 9. Porosity estimates, θ , (a), saturated hydraulic conductivity, K_s , (b) and lateral chloride transport, Cl , (c) of the preferential flowpaths in relation to the dye coverage of different soil horizons as determined from the middle cross-section of the vertically excavated plot (cf. Fig. 3).

1971), and it is possible that they were the reason for dye accumulation at some points in soil at our site as well. In tile-drained clayey fields, earthworm burrows and cracks have been found to have a strong influence on the infiltration processes and formation of subsurface drain flow (e.g., Shipitalo et al., 2004).

4.2. Extent and connectivity of preferential flowpaths

Considering the different elevation levels of the Kangaslampi hillslope (Fig. 1b), the dye has been found to reach the same depth of about 55 cm from the soil surface when poured into unsaturated soil (cf. Backnäs et al., 2015), indicating that this is the lowest depth that preferential flowpaths reach at the Kangaslampi site. At a pine forest site in Slovakia, Brilliant Blue FCF infiltrated to the corresponding depth to sand soil (Homolák et al., 2009), but at the Norway spruce site in South-East Germany, dye was found to infiltrate deeper in the sandy to loamy till (Bogner et al., 2010).

In Kangaslampi, the dye spread most uniformly in the soil material of the E horizon in the midslope area. This indicated that the E horizon was loose and contained a well-connected network of flow routes. The thickness of the E horizon showed considerable variability due to stones: The larger the stones in the E horizon, the thicker the E horizon and the deeper the dye infiltration. This has resulted from the increased infiltration through preferential flowpaths and enhanced podzolization due to dissolution of metal sesquioxides from preferential flowpaths as detected by Backnäs et al. (2012) and Bogner et al. (2012) on Podzolic forest soils. In the B horizon, the dye pattern was patchy, and the last traces of dye were found in the BC horizon. In Fennoscandian tills in general, the hydraulic conductivity, as well as the fraction of preferential flowpaths, has been considered to be low in subsoil, but in the uppermost soil horizons near the soil surface preferential flowpaths have a decisive influence on subsurface flow and transport processes (cf. Lind and Lundin, 1990). Thus, even if a minor volume of preferential flowpaths does exist deep in the C horizon, these flowpaths are not well-connected and may not form a continuous pore network that would enable rapid lateral flow.

Several factors explain the infiltration depth observed in our experiments. First, coarse-textured soils usually wet up fully (Aubertin, 1971). This means that the dye solution poured into the soil may have saturated the soil matrix in the E, B and BC horizons through preferential flowpaths first, before reaching the C horizon. As the volume of dye solution was about a half of the empty pore volume of the E, B and BC horizons, it is possible that a larger volume of dye solution could have led to a deeper dye infiltration than what was observed now. Second, sorption of the blue dye to soil material has been found to increase with increasing background ionic strengths (e.g., Germán-Heins and Flury, 2000). Thus, it is possible that the B horizon adsorbed dye due to its geochemical properties when the dye solution infiltrated down through this layer. Third, earlier studies (Laine-Kaulio et al., 2014a; Laine-Kaulio, 2011), as well as the results of the present study, indicate

that the macropore fraction and the hydraulic conductivity of soil are small in the C horizon, and the lack of preferential flowpaths prevented the dye from reaching the C horizon.

Similarly to the dye pattern in unsaturated soil, heterogeneity in the soil structure and texture resulted in an uneven dye pattern also in saturated soil. Despite the greater adsorption capacity, the high soil bulk density and the strongly cemented Ortstein lenses of the B horizon, the lateral flow experiment revealed clear traces of dye as far as 1.2 m downslope from the dye source when applied to soil that was saturated up to the B horizon. Dye detected in the E horizon at the same distance indicated that lateral flow also occurred to some extent in the highly but not fully saturated E horizon above. This finding is particularly important considering the fact that a full saturation of soil at the site, up to the top of the mineral soil profile, is unlikely in natural conditions. This is due to the high hydraulic conductivity of soil near the soil surface because of the preferential flowpaths, and the low rainfall intensities and amounts of the area; the long-term mean annual precipitation is only 564 mm in Kangaslampi, and the probability of, e.g., a rainfall event of 40 mm h⁻¹ is only 10% in Finland (e.g., Piirainen et al., 2007; Climateguide.fi, 2015).

The lateral dye continuity of over 1.2 m showed that preferential flowpaths self-organized into larger preferential flow systems when the soil saturated. This was despite the fact that individual macropore segments have been found to be less than 50 cm in length (Sidle et al., 2001). It is important to notice that increased ion concentrations decrease the mobility of Acid Blue 9 (Germán-Heins and Flury, 2000). It is therefore possible that the chloride irrigations performed at the same spot one year earlier (Laine-Kaulio, 2011) have affected the lateral transport of the dye. The mean lateral transport velocity of the chloride pulse was as much as 3 m h⁻¹ in initially moist soil during intensive irrigation. To conclude, the dye and chloride experiments together indicated that a continuous lateral network of preferential flowpaths may cover the entire length of the Kangaslampi hillslope when the soil is wet.

4.3. Runoff generation at the Kangaslampi hillslope

According to the available, detailed dual-permeability simulations of lateral chloride transport in the midslope area (Laine-Kaulio et al., 2014a), preferential flowpaths can deliver 140 times more chloride downslope than the soil matrix under highly saturated soil moisture conditions. Compared to the C horizon with a minor preferential flow fraction, the lateral transport capacity of the preferential flow domain was 5-fold in the BC horizon, 50-fold in the B horizon and 76-fold in the E horizon (cf. Fig. 9c). According to Laine-Kaulio et al. (2014b), runoff generation in the midslope area is characterized by the transmissivity feedback phenomenon and dominated by lateral by-pass flow; correlations found between the dye coverage and different characteristics of lateral preferential flow (Fig. 9) support this conception.

Within the shallow soil profiles of the slope shoulder and slope foot study plots, the porosities of preferential flowpaths did not show a reduction with depth, implying that the hydraulic conductivity also does not reduce within the shallow soil depth, and that the transmissivity feedback phenomenon does not apply. Findings of Backnäs et al. (2015) support this conception as they found considerable amount of dye throughout the soil profile at the slope shoulder and slope foot study plots in their experiments. Backnäs et al. (2015) reported also that the E horizon was not fully colored, but instead, the entire soil profiles had distinctly colored and colorless spots. Combining the findings of Backnäs et al. (2015) to our porosity estimates of preferential flowpaths we inferred that the flowpath network in the entire shallow and loose soil profile, as well as flowpaths on the underlying bedrock surface, are crucial for runoff generation at the slope shoulder. At the slope foot, preferential flowpaths in soil below the E horizon seem to be the major runoff sources.

Similar findings to those made at the slope shoulder and slope foot study plots have been made in the Hyttiälä research site in Finland. In Hyttiälä, runoff observations from a 10 year period showed that in a sandy till soil column with an average thickness of 50–70 cm, no surface or surface layer runoff occurred in any conditions, and lateral flow generated in the soil layer above the bedrock when the lower soil layers saturated (Ilvesniemi et al., 2010). In addition to the low soil depth, a stone content that increased with depth explained the runoff generation near the bedrock in Hyttiälä. Pipeflow on the bedrock surface has also been found to be the major source of runoff at, for example, the Maimai experimental hillslope in New Zealand, where the soil thickness was about 60 cm, roots reached the bedrock surface, and pipes were found to extend laterally downslope over distances of tens of meters (McDonnell, 1990; Mosley, 1979).

5. Conclusions

Dyes have shown to be valuable tracers in visualizing flow patterns and pathways in the subsurface (Flury and Wai, 2003). Combined with soil analyses and in situ irrigation experiments with other types of tracers, dye tracer studies can also provide quantitative information on subsurface flow processes (Alaoui and Goetz, 2008). Based on the analysis presented in this study, the following conceptualization of flow formation at the Kangaslampi hillslope can be drawn. Preferential flowpaths reach the depth of about 55 cm from the soil surface. When the soil is dry, water infiltrates into soil via networks of large pores and voids that are formed by rooting activities, erosion related to soil water flow, freezing–thawing phenomena, and soil fauna. These factors contribute to the formation of preferential flowpaths together. Roots alter the pore size distribution of soil, and subsurface water flow and soil frost create voids around stone surfaces. As roots accumulate in the vicinity of stones, where also water and solutes accumulate, preferential flowpaths related to roots are connected to the voids around stones.

In the midslope area, where the stone content is highest, the soil profile thickest and the density of soil high below the E horizon, the role of stone surfaces in the formation of preferential flowpaths is emphasized. At the slope shoulder and slope foot, where the stone content is lower and the soil profiles shallow, large pores and soil pipes caused by roots act most clearly as preferential flowpaths. In addition, the bedrock surface provides a lateral preferential flow route at the slope shoulder where the soil profile is most shallow. The individual preferential flowpaths self-organize into large preferential flow networks that enable fast lateral subsurface flow downslope in highly saturated soil. However, the dominant flow mechanisms behind the runoff generation in different parts of the hillslope are not the same because the soil properties, including the characteristics of the preferential flow networks, are different.

In the midslope area the runoff generation is characterized by lateral by-pass flow and the transmissivity feedback phenomenon, while at the

slope shoulder, the entire soil profile, as well as preferential flowpaths on the underlying bedrock surface, are crucial for runoff generation. At the slope foot, soil below the E horizon is the major runoff contributor. Despite the differences in the characteristics of preferential flowpaths and in the main runoff contributing soil layers, the soil has to be considered two separate, but connected pore domains along the entire hillslope at least in the uppermost 55 cm of the soil profile where preferential flowpaths control the vertical infiltration, as well as the lateral flow downslope.

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